



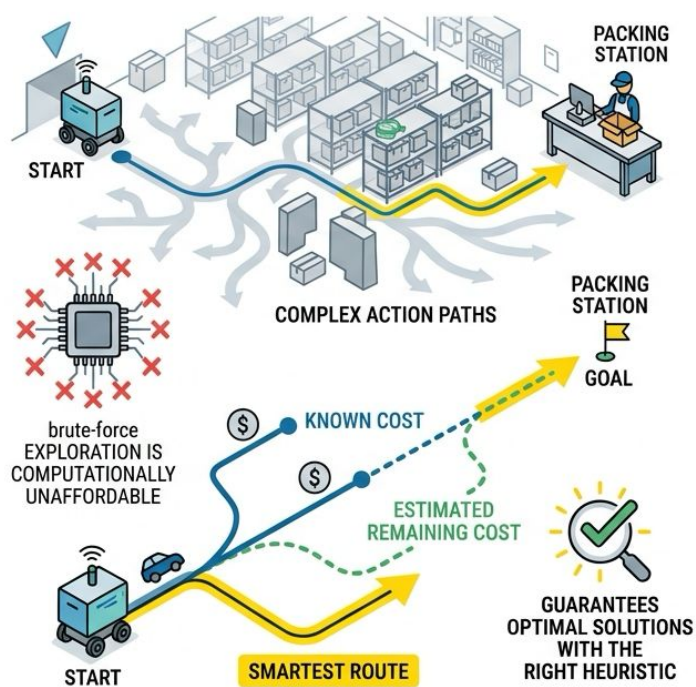
U.S. General Services
Administration



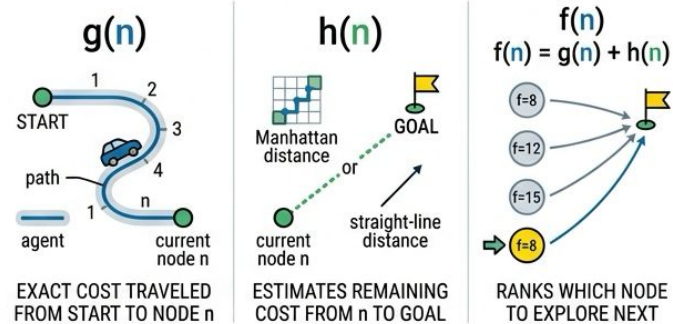
Chapter 5.6: A* Search and Replanning

US General Service Administration
AI Community of Practice - March 2026

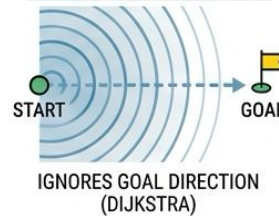
The A* Evaluation Function



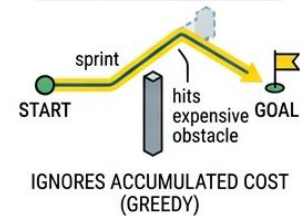
$$f(n) = g(n) + h(n)$$



EXPANDING ONLY BY $g(n)$



EXPANDING ONLY BY $h(n)$

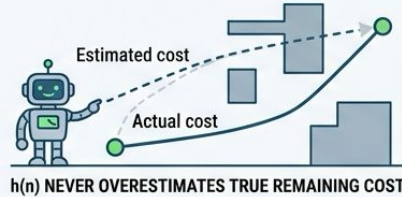


Heuristics & Tradeoffs

Admissible & Consistent Heuristics



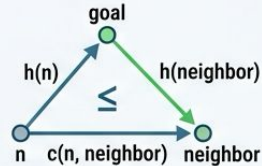
Admissible Heuristics:
 $h(n)$ NEVER OVERESTIMATES
TRUE REMAINING COST



Admissibility Guarantees A* Finds Optimal Path



Admissibility Guarantees
A* Finds Optimal Path



**Consistency: $h(n)$ Respects
Triangle Inequality**



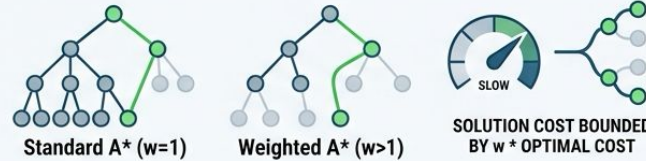
Consistent Heuristics
Prevent Reopening Closed Nodes



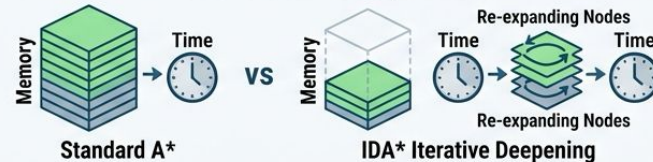
Weighted A* & Practical Tradeoffs

$$f(n) = g(n) + w * h(n) \quad w > 1$$

Higher w , Faster Search, Suboptimal Paths

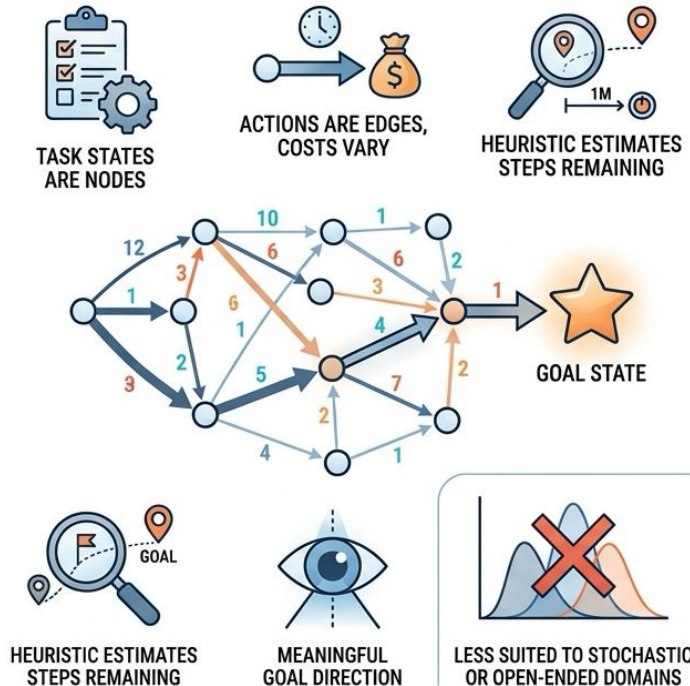


IDA* Trades Memory for Time

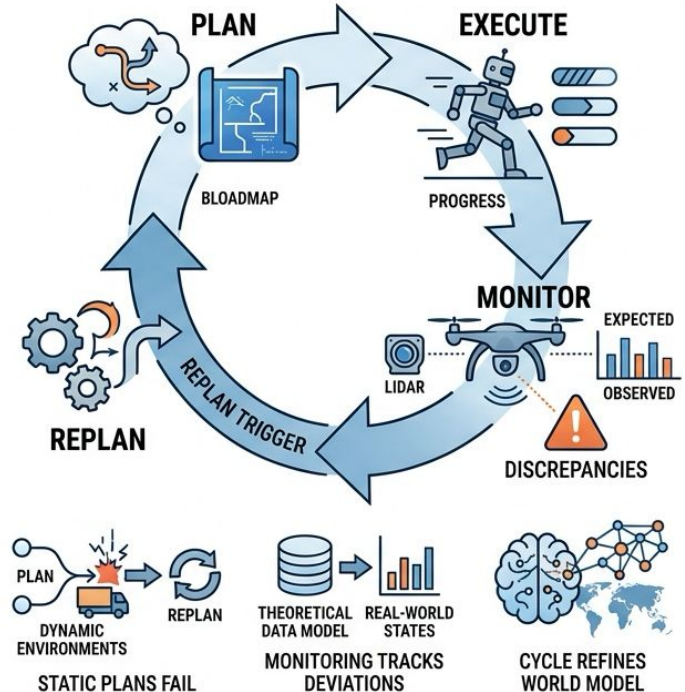


The Plan-Execute-Monitor-Replan Cycle

[A* Beyond Pathfinding -- Agent Task Planning]



[The Plan-Execute-Monitor-Replan Cycle]



Replanning Triggers

ACTION EXECUTION FAILURES



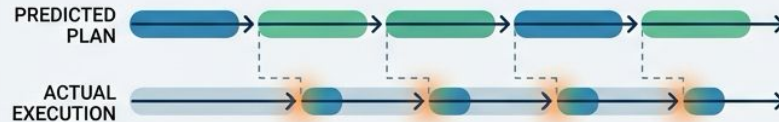
SENSOR NOISE AND ACCUMULATED DRIFT



EXOGENOUS EVENTS (WORLD CHANGES)



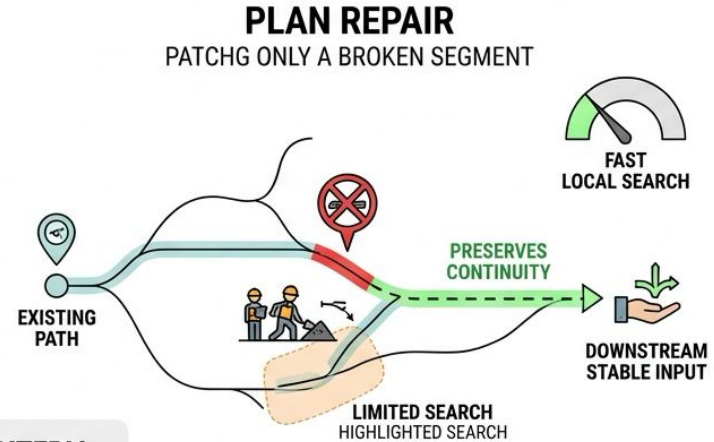
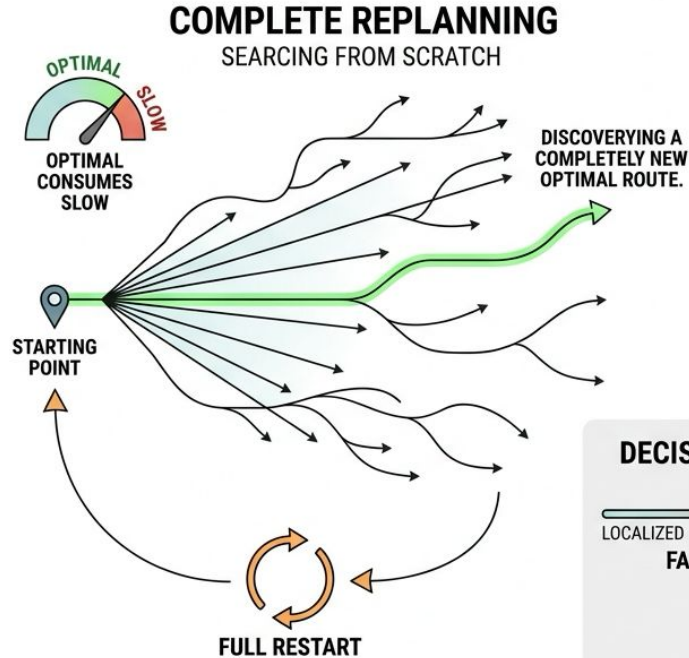
MODEL INACCURACIES & REPEATING PREDICTION ERRORS



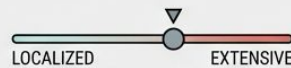
GOAL CHANGES INVALIDATE PURPOSE



Repair vs. Complete Replanning



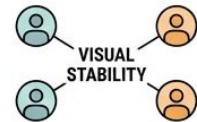
DECISION CRITERIA



FAILURE SCOPE

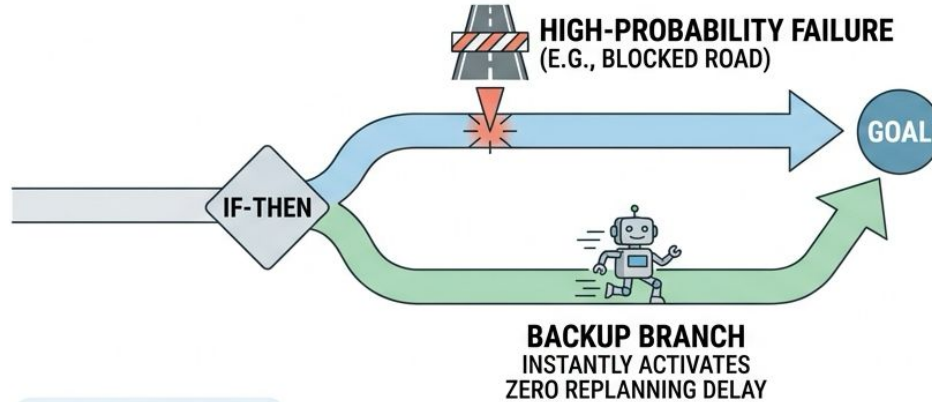


TIME CONSTRAINTS



STABILITY
DOWNSTREAM SYSTEMS, &
MULTI-AGENT COORDINATION

Contingency Planning



WHEN TO USE

P(FAILURE)



PROBABILITY

*

OR

DELAY



TIME

>



PLANNING
COST



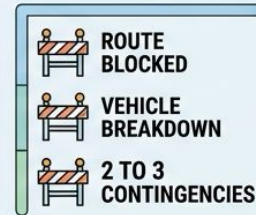
INSTANT ACTIVATION

DIRECT CONNECTION SWITCHING
FROM CONTINGENCY PLAN TO



PAUSE/THINK
ICON

PARETO EFFICIENCY

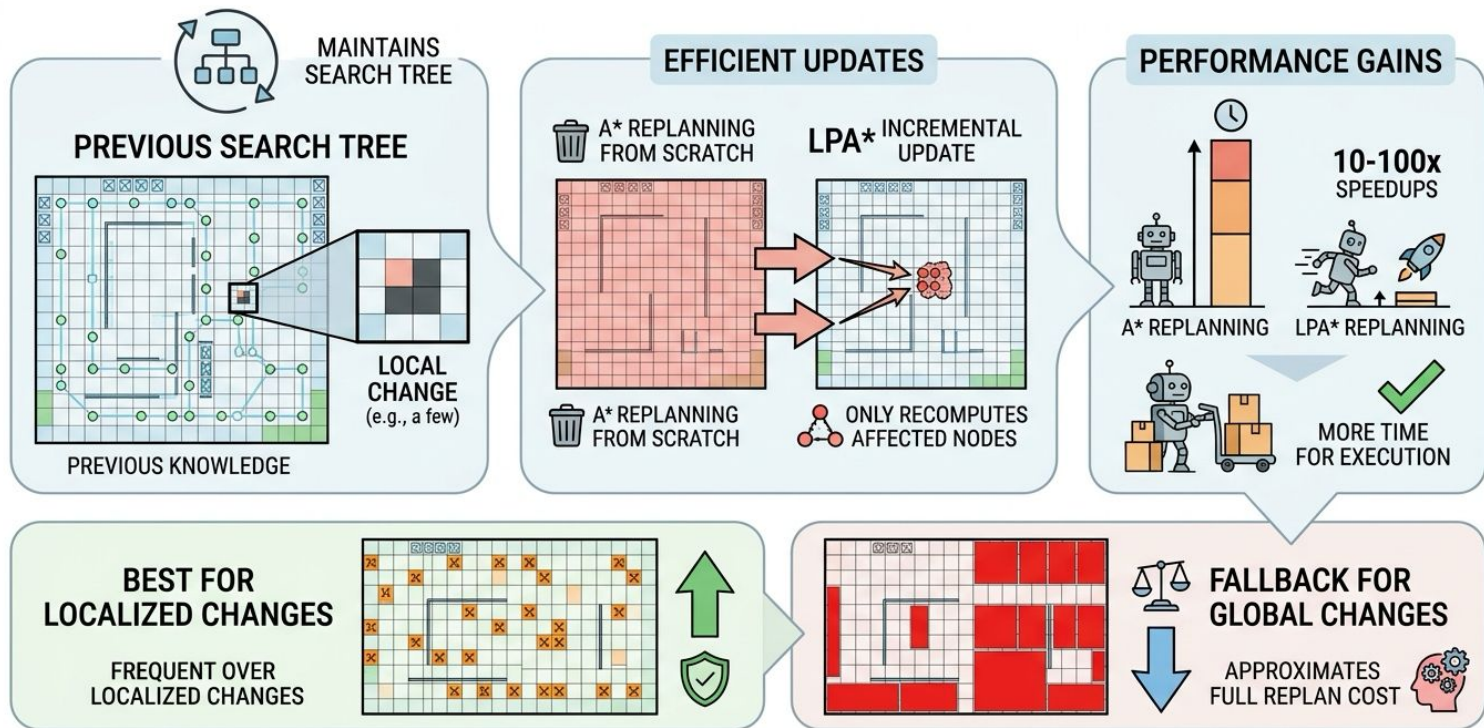


RARE
SURPRISES

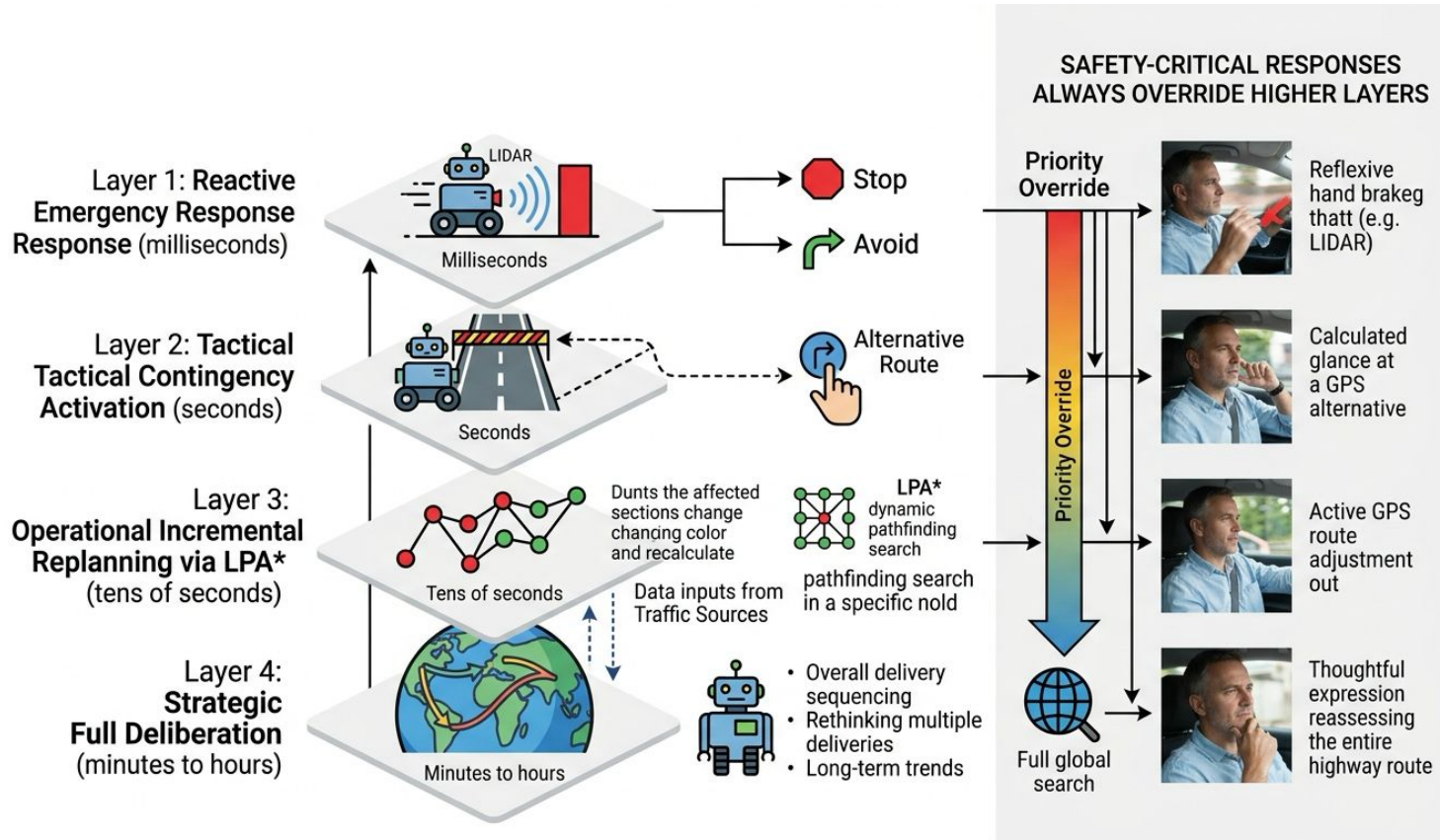


FEW CRITICAL ISSUES ACCOUNT FOR
MAJORITY OF REAL-WORLD DISRUPTIONS.

LPA* -- Incremental Replanning



Multi-Layered Replanning Architecture



Key Takeaways

- A* balances known cost with heuristic estimation for optimal search
 - Admissibility is the mathematical key to optimality guarantees
 - Replanning transforms static plans into adaptive behaviors
 - Choose repair, contingency, or LPA* based on failure characteristics
 - Next chapter: how agents remember past experiences to plan better
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